EXPERIMENT: Write a Prolog Program to print particular bird can fly or not. Incorporate required queries

Aim

To write a **Prolog program to determine whether a particular bird can fly or not** using facts, rules, and logical queries.

Objectives

- To represent bird information using Prolog facts.
- To use rules for determining flying ability.
- To understand Prolog queries and logical inference.
- To handle birds that cannot fly, such as penguins and ostriches.

Algorithm

1. Start the program.
2. Define facts for birds.
3. Define facts for birds that cannot fly.
4. Create a rule can_fly/1.
5. If a bird is a normal bird and is not listed as a flightless bird, conclude that it can fly.
6. Query the program with a particular bird name.
7. Display whether the bird can fly or cannot fly.
8. Stop the program.

CODE

```
birds = [
  "eagle",
  "parrot",
  "pigeon",
  "sparrow",
  "crow",
```

```
    "peacock",
    "penguin",
    "ostrich",
    "kiwi",
    "emu"
]
cannot_fly = [
    "penguin",
    "ostrich",
    "kiwi",
    "emu"
]
def check_bird(bird):
    bird = bird.lower()
    if bird not in birds:
        print(bird, "is not present in the bird database.")
    elif bird in cannot_fly:
        print(bird, "cannot fly.")
    else:
        print(bird, "can fly.")
print("=" * 50)
print("      BIRD FLIGHT CHECKING PROGRAM")
print("=" * 50)
print("\nAvailable birds:")
for bird in birds:
    print("-", bird)
```

```
print("\nBirds that cannot fly:")
for bird in cannot_fly:
    print("-", bird)
print("\n" + "-" * 50)
bird_name = input("Enter a bird name: ")
check_bird(bird_name)
print("-" * 50)
print("Program completed successfully.")
```

OUTPUT

```
IDLE Shell 3.12.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug  6 2024, 20:45:27) [MSC v.1940 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/shaik arshad basha/AppData/Local/Programs/Python/Python312/x
dc.py
==================================================
       BIRD FLIGHT CHECKING PROGRAM
==================================================

Available birds:
- eagle
- parrot
- pigeon
- sparrow
- crow
- peacock
- penguin
- ostrich
- kiwi
- emu

Birds that cannot fly:
- penguin
- ostrich
- kiwi
- emu

--------------------------------------------------
Enter a bird name: peacock
peacock can fly.
--------------------------------------------------
Program completed successfully.
>>> |
                                                              Ln: 32  Col: 0
```

Result

The Prolog program was successfully implemented to determine whether a particular bird can fly or cannot fly. The required queries correctly identify flying birds such as eagle and parrot and flightless birds such as penguin and ostrich.

Conclusion

This program demonstrates the use of Prolog facts, rules, negation, and queries for knowledge-based reasoning. Prolog can automatically determine the flying ability of a bird based on the facts and rules stored in the program.